

## WEI WANG

Email: [wwang93@vols.utk.edu](mailto:wwang93@vols.utk.edu) | [GitHub](#) | [LinkedIn](#)  
Knoxville, TN, USA

### PROFESSIONAL SUMMARY

Ph.D. Candidate in Learning Design and Technology at the University of Tennessee, Knoxville, and co-founder of AskAboutEdu, an evidence-informed decision-support initiative for education. My research examines how generative AI, learning analytics, and evidence-grounded systems can support teaching, learning, critical thinking, AI literacy, and evidence-informed decision making. I use design-based research, mixed methods, and computational methods to build and evaluate educational technologies, including retrieval-augmented generation (RAG) systems and interactive data platforms. I co-developed the AIREA Data Explorer and am first author of a forthcoming CRC Press book on computational analysis of educational data. Previously, I conducted mixed-method evaluations of large-scale smart-education initiatives at iFLYTEK's National Cognitive Intelligence Key Laboratory. Technical expertise includes Python, R, LLM systems, NLP, causal inference, and interactive data visualization.

### EDUCATION

|                                                                                                    |                   |
|----------------------------------------------------------------------------------------------------|-------------------|
| University of Tennessee, TN, USA                                                                   | 08/2023 – present |
| <b>Ph.D. Learning Design and Technology, Education</b>                                             |                   |
| <b>Committee:</b> Dr. Joshua Rosenberg, Dr. Enilda Romero-Hall, Dr. Victor Lee, Dr. Jason Johnston |                   |
| Johnson University, TN, USA                                                                        | 06/2020 – 12/2021 |
| <b>M.A., Education Technology</b>                                                                  |                   |
| Anhui Normal University, Wuhu, China                                                               | 09/2019 – 06/2022 |
| <b>M.S., Education Technology</b>                                                                  |                   |
| Longyan University, Longyan, China                                                                 | 09/2012 – 06/2016 |
| <b>B.S., Software Engineering</b>                                                                  |                   |

### PUBLICATIONS

- **Wang, W.**, Pritchard, C., & Rosenberg, J. M. (accepted for publication, 2026). Analyzing Educational Video Transcripts with LLMs: Accuracy, Named Entities, and Interpretability. *Journal of Educational Data Mining*.
- Duan, Y., Romero-Hall, E., **Wang, W.**, & Morrow, J. A. (under review, 2026). Is Online Learning Really Ready for International Students? Acceptance, Participation, and Cultural Responsiveness in U.S. Higher Education. *Education Sciences*.
- Zhang, H., Rosenberg, J. M., **Wang, W.**, & Sublett, C. (under revision, 2026). How Data Science Has Changed: Evidence from U.S. Job Postings, 2010–2025. *Harvard Data Science Review*.
- Zhang, H., Rosenberg, J. M., Frank, K. A., **Wang, W.**, & Xu, R. (2026). Causal inference and sensitivity analysis methods: A field guide for science education researchers. *OSF Preprints*. [https://doi.org/10.31234/osf.io/kn5qb\\_v2](https://doi.org/10.31234/osf.io/kn5qb_v2)
- Song, Y., **Wang, W.**, Kilgore, C., & Kim, J. (SIGCSE 2027, under review). Developing GenAI Literacy among University Faculty: An Experience Report from a Nationwide Online Workshop. Paper - ERT.
- **Wang, W.**, Rosenberg, J. M., & Koehler, M. J. (2025). Trends in AI Curriculum Materials: Insights from Data Mining a Teacher Marketplace. [EdArXiv Preprint](#).
- Ping, H., **Wang, W.**, Xie, Y., Lv, S., Li, J., & Weng, L. (2025, April). Generative AI Dependency in Higher Education: Investigating Continuance Intention, Cognitive Response, and Creativity. In 2025 7th International Conference on Computer Science and Technologies in Education (CSTE) (pp. 615-621). IEEE. <https://doi.org/10.1109/CSTE64638.2025.11092242>
- Allman, B., Kimmons, R., **Wang, W.** et al. Trends and Topics in Educational Technology, 2024 Edition. *TechTrends* (2024). <https://doi.org/10.1007/s11528-024-00950-5>
- Sarah Narvaiz, Qinyun Lin, Joshua M. Rosenberg, Kenneth A. Frank, Spiro J. Maroulis, **Wei Wang**, Ran Xu (2024). Konfound: An R Sensitivity Analysis Package to Quantify the Robustness of Causal Inferences. *Journal of Open Source Software*, 9(95), 5779, <https://doi.org/10.21105/joss.05779>
- Ye, F.-z., & **Wang, W.** (2023). Returning to the main position of classroom education: A study on the teaching efficacy of smart classrooms and its influencing factors. *Journal of Bingtuan Education Institute*, 33(2), 57-65. (CNKI)

- **Wang, W.**, Duan, Y., Wang, Q., & Liu, H. (2021). An ECM-ISC Based on College Students' Continued Learning Intention toward E-Learning Space Post COVID-19. *Open Journal of Social Sciences*, 9, 377-395. DOI: [10.4236/jss.2021.912026](https://doi.org/10.4236/jss.2021.912026)
- **Wang, W.** (2020). Individual-construction strategy of school informatization based on MOOC. *China Education Technology & Equipment*. (CNKI)

## BOOKS & GUIDEBOOKS

---

**Wang, W.**, Akcaoglu, M., Rosenberg, J. M., & Kellogg, S. (under contract, anticipated publication late 2026). [\*Computational social science cookbook with R: A practical guide\*](#) (a volume in the *Chapman & Hall/CRC Big Data Series*). CRC Press.

Song, Y., Kilgore, C., & **Wang, W.** (Eds.). (2026). *[AI and Me] Generative AI guidebook for higher education faculty*. [EdtechBook].

## RESEARCH EXPERIENCE

---

### Project: AskAboutEdu — Evidence-Informed Decision Support for Education

Role: Co-founder & Researcher

- Designed and deployed AskAboutEdu, an open-source AI interface that connects large language models (LLMs) with structured educational research evidence, enabling users to query intervention reports and receive answers grounded in primary sources.
- Built an end-to-end retrieval-augmented generation (RAG) pipeline using Python, Streamlit, and sentence-level embeddings to index and retrieve evidence from What Works Clearinghouse (WWC) intervention reports.
- Developed a reproducible data processing workflow to extract, normalize, and chunk long-form PDF research reports into structured JSON corpora, preserving document-level metadata for transparent citation and traceability.
- Implemented evidence-constrained prompting strategies that require the model to answer only from retrieved excerpts, reducing hallucination and enabling explicit source attribution at the chunk level.
- Designed the interface to support researchers, policymakers, and educators in exploring educational evidence interactively, serving as a prototype for transparent, explainable, and evidence-based AI systems in applied education research.

### Project: Next Level Teaching Blueprint — Scaling AI-Assisted Critical Thinking Across Research Methods Courses

University of Tennessee | UTK ORIED-funded | PI: Dr. Fatima T. Zahra | Role: Student Principal Investigator & Graduate Researcher

- Co-led a multi-phase design-based research initiative examining how generative AI can function as a structured discourse partner to strengthen higher-order thinking, critical thinking, and AI literacy in undergraduate and graduate research methods courses.
- Contributed to the design and implementation of constraint-first, dialogic learning activities that require students to articulate their initial reasoning, interrogate assumptions through interaction with an LLM, revise their responses, and document metacognitive reflections.
- Developed an integrated mixed-methods research workflow linking pre/post measures of disciplinary learning, AI literacy, and critical thinking with student learning diaries, AI chat logs, assignment artifacts, classroom observations, and faculty feedback across the Spring 2026 CEHHS Phase 1 pilot.
- Conducted quantitative analyses for the Spring 2025 pre-pilot study, including data validation, demographic reporting, pre/post comparisons, and the production of publication-ready tables and visualizations for statistical learning and AI literacy outcomes.
- Designed computational analyses of student–AI discourse, including interaction-sequence modeling, dialogue-act classification, linguistic markers of learner authority and deference, and learner positioning of AI as an answer generator, validator, or co-thinker.
- Collaborated with an interdisciplinary research team and participating faculty to translate empirical findings into scalable pedagogical principles, faculty implementation resources, and scholarly publications, including a Phase 1 manuscript targeted for CHI 2027.

**Project: The Advanced Infrastructure, Energy, and Agriculture (AIREA) Data Explorer (URL: [AIREA Data Explorer](#))**

University of Tennessee | Funded by JP Morgan Chase

PI: Dr. Cameron Sublett | [Project Site](#)

R Shiny Repository: <https://github.com/data-edu/airea-data-explorer-shiny>

Data Prep Repository: <https://github.com/data-edu/airea-data-explorer-prep>

- Designed and deployed the AIREA Data Explorer, a web-based visualization platform showing how U.S. community colleges prepare students for high-opportunity occupations in infrastructure, energy, agriculture, and related sectors.
- Built the platform using R Shiny and Mapbox GL JS, integrating national datasets to visualize changes in job opportunities and credential completions since 2010.
- Collaborated with researchers at the Community College Research Center (Columbia University) to ensure research rigor, data accessibility, and user-centered interface design.
- The dashboard allows policymakers, educators, and researchers to analyze alignment between community college programs and labor market needs, promoting data-driven educational planning.
- Serves as a model for open, reproducible data communication in applied educational research and workforce analytics.

**Institute of Education Sciences (IES) Funded Project — Quantifying the Robustness of Causal Inferences**

Award No. R305D220022 | Michigan State University (Collaborative)

PI: Dr. Kenneth Frank | [IES Project Info](#)

- Contributed to the development and validation of the “[Konfound](#)” R package, an open-source statistical tool for causal inference robustness analysis.
- Collaborated with cross-institutional teams to test algorithms, ensure code reproducibility, and co-author peer-reviewed publications in *Journal of Open Source Software (JOSS)*.
- Enhanced model evaluation procedures relevant to AI-based educational analytics and scalable statistical workflows.

**Project: Individualized Teaching and Educational Governance Supported by Artificial Intelligence**

iFLYTEK Education Technology Institute | Bengbu, China | Grant No. IED2020-Z003

PI: Dr. Bang Qi Liu

- Conducted field studies on AI-supported educational governance and school-level adoption of smart classroom technologies.
- Designed and implemented qualitative interview protocols for teachers and principals, then coded and analyzed data to assess the impact of AI-driven teaching models.
- Authored regional governance reports that informed provincial education policy and smart classroom development strategies.

**Project: Anhui Philosophy and Social Science Planning Project — Knowledge Supply and Deep Learning in the “Internet+” Era.**

Anhui Normal University | Grant No. AHSKY2019D036

PI: Dr. Hehai Liu

- Developed and validated a survey instrument grounded in the ECM-ISC model to measure students’ continued engagement with online learning environments.
- Applied multilevel regression and confirmatory factor analysis using R and SPSS to assess behavioral intention and technology adoption factors.
- Published as first author in peer-reviewed journals, contributing to research on digital learning adoption and instructional technology design.

## UNIVERSITY TEACHING, CURRICULUM, AND FACULTY DEVELOPMENT

---

### Teaching Assistant, ESM 250: Applied Statistical Concepts for Decision Makers

*Undergraduate course | Instructor: Fatima T. Zahra | University of Tennessee, Knoxville | Spring 2025*

- Facilitated weekly applied sessions in R, Excel, Stata, and generative AI, guiding debugging, method selection, and a statistical consulting project.
- Developed instructor-reviewed model solutions, maintained Canvas discussions, and reinforced ethical GenAI use, independent problem solving, critical evaluation, and attribution.
- Supported classroom research through survey and assessment administration, documentation of GenAI engagement, and FERPA-aware handling of student interaction data.

### Instructor & Project Contributor, AI and Me: Navigating GenAI for Higher Education Faculty

*CEHHS and Teaching & Learning Innovation | University of Tennessee, Knoxville | 2025–Present*

- Supported design and delivery of a five-session national professional-development series on responsible GenAI use in teaching, assessment, research, media creation, and faculty workflows.
- Developed the self-paced Canvas course, organized accessible transcripts and resources, supported completion certificates, and co-edited the companion EdTech Books guidebook.

### Faculty Development Facilitator, Teaching with Generative AI

*Digital Learning Unit | University of Tennessee, Knoxville | Apr 2024–Present*

- Facilitate professional learning on GenAI foundations, course integration, assessment, ethics, privacy, and responsible practice; provide formative feedback on faculty implementation plans.
- Analyze participant surveys, interviews, and reflections to refine curriculum and support for faculty across disciplinary and technical backgrounds.

### Instructor & Adaptive Learning Specialist

*Shanghai Squirrel AI Education Technology Co., Ltd. | Dec 2017–Aug 2019*

- Implemented adaptive learning workflows for K–12 learners, used diagnostic data to personalize learning pathways, and trained teachers to integrate the system into instruction.

## WORK EXPERIENCE

---

### Digital Learning Unit, University of Tennessee, Knoxville, TN, USA

Teaching Assistant & Research Assistant | Aug 2023 – Present

- Conducted research on AI-enhanced online teaching practices, synthesizing empirical evidence to inform design recommendations for higher-education contexts.
- Designed and administered survey and interview protocols; led qualitative coding and mixed-method analysis on faculty perceptions and adoption of AI tools.
- Collaborated with a research team to prototype AI-informed instructional frameworks (e.g., teacher support workflows and reusable design patterns) and document implementation guidance.

### National Cognitive Intelligence Key Laboratory, Institute of Educational Technology, iFLYTEK, China

Researcher | Mar 2021 – Jun 2023

- Conducted research in Smart Education Demonstration Zones, evaluating AI-enabled learning platforms and K–12 smart classroom ecosystems across regional pilots.
- Led mixed-method evaluation studies to examine impacts on teaching practices and student engagement (learning analytics + surveys/interviews/classroom observations).
- Maintained an intelligent-education policy and initiative database supporting evidence-based reporting and consultation for local education authorities.
- Contributed to the China Annual Blue Book of Artificial Intelligence Education Development and coordinated research-practice communication through the Annual Intelligent Education Development Forum, engaging researchers, educators, and product teams.

## CONFERENCE PRESENTATIONS

---

- **Wang, W** (2026). Localized Large Language Models for Educational Research: Two Privacy-Preserving Case Studies in Vision and Text Analysis. Chicago, AECT 2026.
- **Wang, W** (2026). Bridging Workforce Data and Decision-Making with a RAG-Enabled Community College Data Explorer. Stanford, Stanford EDS 2026.
- **Wang, W** (2026). Developing Prompt Engineering Competence Framework: Comparing China, EU, and U.S. Educators' Digital Literacy Standards. Los Angeles, AERA 2026.
- **Wang, W** (2026). Educational Video Transcript Analysis with LLMs: Improving Entity Recognition and Qualitative Insights. Los Angeles, AERA 2026.
- **Wang, W** (2025). Bridging the AI Gap: Collaborative Strategies for Effective Integration in Higher Education. Orlando, OLC 2025.
- **Wang, W** (2025). Aligning Degree Production and In-Demand Middle Skill Occupations: National and Regional Insights. Seattle, APPAM 2025.
- **Wang, W** (2025). Faculty Voices: Sentiment and Thematic Analysis of GenAI Integration in Higher Education Course Design. Las Vegas, AECT 2025.
- Hongjie Ping, **Wang, W** (2025). Generative AI Dependency in Higher Education: Investigating Continuance Intention, Cognitive Response, and Creativity. The 7th International Conference on Computer Science and Technologies in Education (CSTE). Wuhan, China. CSTE 2025.
- **Wang, W** (2025). Exploring Dimensions of AI Integration and Designing an Effective GenAI-Based Teaching Model. Denver, AERA 2025.
- **Wang, W.** (2024). Navigating Sentimental Landscapes: An In-depth Analysis of ChatGPT Implementation in Universities through Policy Examination. Philadelphia, AERA 2024.
- **Wang, W.** (2024). Implementing Generative AI in Practice: Designing Assessments and Learning Activities in Faculty Development Program. Kansas City, AECT 2024.

---

## SERVICE

Peer Reviewer, Journal of Research on Technology in Education, 2024–2025.  
Peer Reviewer, Journal of Statistics and Data Science Education, 2025.  
Peer Reviewer, AECT 2025.  
**President** of CSSA, Student Union, University of Tennessee.

---

## PROFESSIONAL SKILLS & RECOGNITIONS

### TECHNICAL & ANALYTICAL SKILLS

- Programming Languages: Python, R
- AI/ML Frameworks: PyTorch, Hugging Face, OpenAI API, Scikit-learn
- Statistical & Modeling Tools: RStudio, SPSS, Mplus, Amos
- Data Science: Educational data mining, LLM evaluation, causal inference, NLP applications
- Data Visualization: R Shiny, Mapbox GL JS, Tableau, ggplot2
- Version Control & Cloud: Git/GitHub, AWS, Google Cloud
- Web & App Development: HTML/CSS, JavaScript, and mobile/web educational platforms

### RESEARCH & DESIGN SKILLS

- Instructional Design: Design-based research for AI-supported teaching and learning
- Empirical Research: Mixed methods, survey design, factor analysis, and regression modeling
- AI for Education: LLM and GenAI integration in classroom and faculty-development contexts
- Evaluation & Assessment: Quantitative and qualitative evaluation of educational technologies and instruction
- Publication & Dissemination: Peer-reviewed articles, technical reports, and public data dashboards

## **CERTIFICATIONS**

- National English Teacher Qualification Certificate (Junior High, China)
- National Mathematics Teacher Qualification Certificate (Junior High, China)
- National IT Teacher Qualification Certificate (Senior High, China)

## **AWARDS & HONORS**

- Julius D. McElroy Scholarship, University of Tennessee, 2023
- GrowSTEM Project Fund Scholarship, University of Tennessee, 2023
- Outstanding Master's Thesis Award, Anhui Normal University, 2022